

华中科技大学 物理实验报告

学生姓名: _____ 学号: _____ 成绩: _____

实验时间: 第 _____ 周, 星期 _____, 上午, 下午, 晚上, (H "V") 教师签字: _____

报告柜号: _____

实验房间号: _____

实验名称: 参考卷

实验编号: _____

1. 假设两绝热线相交于A点, 作一等温线分别与两绝热线交于B、C两点。

考虑 C→B→A→C 循环, 热机从单一热源吸热, 并对外做功, 而并未引起任何变化, 这违反了热力学第二定律开尔文表述, 故不可能。故 T-V 图上两绝热线不能相交。

2. 由热力学第一定律, $\Delta U = Q + W$, 内能变化的原因是①做功 ②传热

$dU = \sum \alpha_i d\epsilon_i + \sum \epsilon_i d\alpha_i$, 前者对应做功, 后者对应传热。

3. 氦气为单原子分子, $\gamma = \frac{5}{3}$, 空气近似为双原子分子, $\gamma = \frac{7}{5}$, $\gamma_{\text{空}} > \gamma_{\text{氦}}$ 。

而绝热过程中, $\alpha = \frac{V}{V_0} T^{\frac{\gamma}{\gamma-1}} = W$, 做功W相同时, 由于 $\gamma_{\text{空}} > \gamma_{\text{氦}}$, 氦气对应的T更大。

故温度比空气升温快。

4. 当温度 T < 特征温度 Θ 时, $e^{-\frac{\epsilon}{kT}} \rightarrow 0$, 粒子几乎全部处于基态, 该自由度的热力学对热容贡献可忽略不计, 视为自由度冻结。

如 T 一般情况下 $\Theta_{\text{平}} \gg \Theta_{\text{转}} \gg \Theta_{\text{振}}$, 较自由度的比热容随温度升高而增加。

5. 对于理想气体, 一个量子的能量是不受限制的。

$N = N(0) + \int_0^{\infty} \frac{V \omega^3}{(2\pi)^3} \frac{d\omega}{e^{\frac{\hbar\omega}{kT}} - 1}$, 当 T 足够高时, 量子效应可忽略, $N \approx N(0)$ 。

当 T 足够低时, $T < T_c$ 时, 玻色子将全部凝聚于基态, $N \approx N(0)$ 。

6. 不会。对于光子气体而言, 光子数 N 可变, $\mu = 0$ 。

另一方面, $N = \int_0^{\infty} \frac{V \omega^3}{(2\pi)^3} \frac{d\omega}{e^{\frac{\hbar\omega}{kT}} - 1} = \frac{8\pi V}{(2\pi)^3} \int_0^{\infty} \frac{\omega^3}{e^{\frac{\hbar\omega}{kT}} - 1} d\omega$, 可见 $N \propto T^3$ 。

当 T → 0 时, $N \rightarrow 0$, 光子数趋于 0, 不可能发生玻色凝聚。

7. 由于电子属于费米子, 服从费米-狄拉克分布, 服从泡利不相容原理。

当 T = 0 K 时, 电子将占据最低的能量向上排布, 直至填满为止, 此时在最高占据的能量 μ_0 。

$f = \begin{cases} 1, & \epsilon \leq \mu_0 \\ 0, & \epsilon > \mu_0 \end{cases}$ 所有能量 $\epsilon < \mu_0$ 的量子态都被电子所占据, 而 $\epsilon > \mu_0$ 则空出。

8. 在化简 μ_0 上投影电子的相量为

$$f(\mu_0) = \frac{1}{e^{\frac{\mu_0}{kT}} + 1}$$

μ_0 以上没有电子的相量为

$$1 - f(\mu_0) = \frac{1}{e^{\frac{\mu_0}{kT}} + 1} = \frac{1}{e^{\frac{\mu_0}{kT}} + 1}$$

二者相等, 证毕。

$$\text{二. (1) } dQ = C_m dT, dS = \frac{C_m dT}{T}$$

设两物体从 T_1 加热至 T_2 的熵变 $\Delta S = \int_{T_1}^{T_2} \frac{C_m dT}{T} = C_m \ln \frac{T_2}{T_1}$

若系统熵变 $\Delta S = \Delta S_1 + \Delta S_2 = \int_{T_1}^{T_2} \frac{C_m dT}{T} + \int_{T_2}^{T_1} \frac{C_m dT}{T} = 0$

由热力学第二定律, $\Delta S \geq 0$, 故 $T_2 \geq T_1$, $T_2 = T_1$ 。

而 $Q = C_m (T_2 - T_1) = C_m (T_2 - T_1)$

$= C_m (T_2 + T_1 - 2T_1)$

$= C_m (T_2 - T_1)$

三. (1) $E = \frac{1}{2} m v^2 = \frac{1}{2} m \frac{h^2 k^2}{m^2} = \frac{\hbar^2 k^2}{2m}$

$\bar{E} = \int_0^{\infty} E g(E) f(E) dE = \int_0^{\infty} \frac{\hbar^2 k^2}{2m} \frac{4\pi k^2 dk}{(2\pi)^3} \frac{1}{e^{\frac{\hbar^2 k^2}{2m kT}} + 1} dE$

$= \frac{1}{\pi^2} \int_0^{\infty} \frac{k^4 dk}{e^{\frac{\hbar^2 k^2}{2m kT}} + 1} = \frac{1}{\pi^2} \int_0^{\infty} \frac{k^4 dk}{e^{\frac{\hbar^2 k^2}{2m kT}} + 1}$

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$$\text{五. (1) } N = \int_0^{\infty} \frac{V \omega^3}{(2\pi)^3} \frac{d\omega}{e^{\frac{\hbar\omega}{kT}} - 1} \Rightarrow P_F = \hbar \left(\frac{3N}{4\pi} \right)^{\frac{1}{3}}$$

$$\mu_0 = \frac{\hbar^2 k_F^2}{2m} = \frac{\hbar^2}{2m} \left(\frac{3N}{4\pi} \right)^{\frac{2}{3}}$$

$$U = \int_0^{\infty} \frac{V \omega^3}{(2\pi)^3} \frac{\hbar\omega}{e^{\frac{\hbar\omega}{kT}} - 1} d\omega = \frac{4\pi V}{(2\pi)^3} \int_0^{\infty} \frac{\omega^4}{e^{\frac{\hbar\omega}{kT}} - 1} d\omega = \frac{3}{8} N \mu_0$$

$$P = \frac{2U}{3V} = \frac{2}{3} \mu_0$$

$$\Rightarrow P = \frac{2}{3} \mu_0 = \frac{2}{3} \frac{\hbar^2}{2m} \left(\frac{3N}{4\pi} \right)^{\frac{2}{3}} = K N^{\frac{2}{3}} \Rightarrow V \propto N^{\frac{2}{3}}$$

$$\left(\frac{V}{N} \right)^{\frac{3}{2}} = \frac{3V}{2P} = \frac{3}{2} \frac{1}{\mu_0}$$

$$\Rightarrow K_F = \frac{1}{V} \left(\frac{3V}{2P} \right)^{\frac{2}{3}} = \frac{3}{2} \frac{1}{\mu_0}$$

2022 年

1. $\Delta U = Q + W$, 内能变化的原因是做功和传热, $dU = \sum \alpha_i d\epsilon_i + \sum \epsilon_i d\alpha_i$ 。

2. 绝热自由膨胀, $Q = 0$, $W = 0$, $\Delta U = 0$, 温度不变。

3. 绝热自由膨胀, $Q = 0$,